EMS Optimization: End-to-End Notebook Output Comparison Report

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Purpose: Audit what the end-to-end notebook (`01_end_to_end_workflow.ipynb`) produces vs. what exists across the full project.

Section A: What the End-to-End Notebook Currently Produces

Visualizations (8 inline plots, 0 saved to disk)

#	Section	Visualization	Type
1	3.2 Temporal demand patterns	Hourly Crash Rate + Day-of-Week Pattern (2-panel)	Bar charts
2	3.3 Spatial demand distribution	Precinct-Level Demand Rates	Horizontal bar chart
3	3.4 Distance matrix & coverage	Distance Matrix heatmap	Heatmap
4	3.4 Distance matrix & coverage	Nearest-Firehouse Distance per Precinct	Bar chart
5	4.2 Policy comparison	Response Time vs Fleet Size + Coverage vs Fleet Size (2-panel)	Line charts
6	4.3 Trade-off	Response Time vs Coverage Trade-off	Scatter plot
7	6.1 Simulation results	P0 vs P2 Response Time (boxplot)	Box plot
8	6.2 Fleet sensitivity	Fleet Size Sensitivity — P2 Policy	Line chart
9	6.3 Demand sensitivity	Demand Intensity Sensitivity — P2 Policy	Line chart

Note: No figures are saved to `results/figures/` — all are `plt.show()` only.

Tables / Summary Statistics (6 printed outputs)

#	Section	Output	Format
1	1. Setup	Configuration summary (K values, capacity, tau, horizon, reps, seed)	Printed text
2	3.1 Dataset overview	Key Statistics (total crashes, date range, duration, rates, counts)	Printed text
3	4.1 Optimization results	Expected Response Time pivot table (policy × K)	Printed DataFrame
4	4.1 Optimization results	Demand Coverage pivot table (policy × K)	Printed DataFrame
5	5.3 Validation summary	Verification (4 tests) + Validation (3 pilots) pass/fail summary	Printed text
6	7. Summary	Full results table (K, policy, RT, coverage, stations_used) + best policy per K + key findings	Printed text

Verification & Validation Tests (7 tests, all inline)

- 4 verification tests (toy example, zero demand, single-unit saturation, extreme demand stress)
- 3 validation pilots (P0 vs P2, fleet sensitivity, demand sensitivity)

Section B: What Exists in the Full Project

Figures in `results/figures/` (60 PNG files)

EDA & Demand Visualization (14 figures)

- `fig_crash_heatmap.png` — Crash density heatmap
- `fig_daily_demand.png` — Daily demand time series
- `fig_hourly_demand.png` — Hourly demand distribution
- `fig_hourly_rates.png` — Hourly arrival rates

- `fig_temporal_trends.png` — Temporal trend decomposition
- `fig_firehouses_map.png` — Firehouse locations map
- `fig_precinct_demand.png` — Precinct demand chart
- `fig_precinct_density.png` — Precinct density visualization
- `fig_demand_model_fit.png` — Input model goodness-of-fit
- `precinct_demand_heatmap.png` — Precinct-level demand heatmap
- `precinct_demand_rates_improved.png` — Improved precinct demand rates chart
- `seasonal_decomposition.png` — Seasonal decomposition plot
- `seasonal_heatmap.png` — Seasonal heatmap
- `seasonal_patterns.png` — Seasonal patterns overview

Service & Travel Model (4 figures)

- `distance_matrix_heatmap.png` — Distance matrix heatmap
- `service_time_distribution.png` — Service time distribution
- `tod_speed_factors.png` — Time-of-day speed factors
- `travel_time_by_tod.png` — Travel time by time of day
- `nhpp_arrivals_demo.png` — NHPP arrival process demo

Optimization Results (8 figures)

- `fig_policy_comparison.png` — Policy comparison summary
- `fig_tradeoff_curve.png` — Response time vs coverage trade-off
- `opt_allocation_comparison.png` — Allocation comparison across policies
- `opt_inputs.png` — Optimization input visualization
- `opt_sensitivity.png` — Optimization sensitivity analysis
- `policy_comparison_panel_K20_cap2.png` — Policy comparison panel at K=20, cap=2
- `optimization/allocation_heatmaps.png` — Allocation heatmaps by policy
- `optimization/policy_comparison.png` — Policy comparison detail

Production Experiment Results (4 figures)

- `exp1_policy_comparison.png` — Experiment 1: baseline policy comparison
- `exp2_fleet_sensitivity.png` — Experiment 2: fleet size sensitivity
- `exp3_demand_sensitivity.png` — Experiment 3: demand scaling sensitivity
- `exp4_service_robustness.png` — Experiment 4: service time robustness

Publication-Quality Figures (5 figures)

- `pub_fig1_policy_comparison.png` — Publication: Policy comparison
- `pub_fig2_fleet_sensitivity.png` — Publication: Fleet sensitivity
- `pub_fig3_demand_robustness.png` — Publication: Demand robustness
- `pub_fig4_service_sensitivity.png` — Publication: Service sensitivity
- `pub_fig5_performance_heatmap.png` — Publication: Performance heatmap

Verification & Validation (3 figures)

- `validation_p0_vs_p2.png` — Validation: P0 vs P2 comparison
- `validation_sensitivity_K.png` — Validation: fleet sensitivity
- `validation_sensitivity_demand.png` — Validation: demand sensitivity
- `verification_toy_timeline.png` — Verification: toy example event trace

Capacity Sensitivity (2 figures)

- `capacity_sensitivity_heatmap.png` — Capacity sensitivity heatmap (P0/P1/P2 × cap=1–5 at K=20 and K=40)
- `firehouse_capacity_analysis.png` — Firehouse capacity analysis

CBD / Robustness Analysis (6 figures)

- `cbd_equity_tradeoff_summary.png` — CBD equity trade-off
- `cbd_heatmap.png` — CBD demand heatmap
- `cbd_response_comparison.png` — CBD response time comparison
- `cbd_robustness_enhanced.png` — CBD robustness enhanced view
- `cbd_scenario_comparison.png` — CBD scenario comparison
- `fig_cbd_comparison.png` — CBD vs non-CBD comparison

Queue & Advanced Analysis (4 figures)

- `queue_comparison_by_policy.png` — Queue comparison by policy
- `queue_heatmap.png` — Queue heatmap
- `queue_vs_demand.png` — Queue vs demand
- `queue_vs_fleet_size.png` — Queue vs fleet size

P0 Spatial Analysis (3 figures)

- `p0_spatial_map.png` — P0 spatial allocation map
- `p0_spatial_metrics.png` — P0 spatial metrics
- `p0_spatial_north_south.png` — P0 north/south comparison

Fleet & Trade-off (3 figures)

- `fleet_sensitivity_dual.png` — Fleet sensitivity dual panel
- `response_time_coverage_tradeoff_improved.png` — Improved trade-off curve
- `response_time_coverage_tradeoff_zoomed.png` — Zoomed trade-off curve
- `response_time_distribution_by_policy.png` — RT distribution by policy

Summary Dashboard (1 figure)

- `project_summary_dashboard.png` — Project summary dashboard

Tables in `results/tables/` (18 CSV files)

File	Content
`optimization_comparison.csv`	Policy comparison across all K values
`table1_baseline_comparison.csv`	Table 1: Baseline vs optimized comparison
`table2_anova_summary.csv`	Table 2: ANOVA results summary
`table3_pairwise_comparisons.csv`	Table 3: Pairwise policy comparisons
`table4_sensitivity_summary.csv`	Table 4: Sensitivity analysis summary
`descriptive_statistics.csv`	Descriptive statistics by policy & K
`anova_results.csv`	Full ANOVA results
`confidence_intervals.csv`	Confidence intervals for RT by policy
`effect_sizes.csv`	Cohen’s d effect sizes
`posthoc_comparisons.csv`	Tukey HSD post-hoc comparisons
`exp1_summary.csv`	Experiment 1 summary
`exp2_pivot_rt.csv`	Experiment 2 response time pivot
`exp3_pivot_rt.csv`	Experiment 3 response time pivot
`exp4_pivot_rt.csv`	Experiment 4 response time pivot
`sensitivity_summary.csv`	Sensitivity summary
`queue_statistics.csv`	Queue statistics
`queue_anova.csv`	Queue ANOVA results
`cbd_comparison.csv` / `cbd_summary_all.csv`	CBD comparison data
`seasonal_analysis.csv`	Seasonal analysis results

Other Notebooks' Unique Outputs

Notebook	Unique Outputs Not in End-to-End
`02_eda_spatiotemporal`	Crash density heatmap, monthly/seasonal patterns, CBD vs non-CBD breakdown, firehouse overlay map
`03_input_modeling`	Homogeneous Poisson fit, NHPP model diagnostics, goodness-of-fit tests, lambda table export
`04_service_travel_proxy`	Service time distribution, ToD speed factors, NHPP arrival demo, call timeline example
`05_optimization`	Allocation heatmaps by policy, diminishing returns analysis, spatial maps
`06_simulation_debug`	Verification event trace timeline, detailed pilot tables
`07_production_results`	4-experiment panel figures (810 simulation results), combined summary table
`08_statistical_analysis`	ANOVA, Tukey HSD, Cohen's d, confidence intervals, assumption checks
`09_cbd_analysis`	CBD vs non-CBD comparison, CBD scenario analysis, spatial coverage analysis

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Section C: Gap Analysis

Critical Gaps (key results that should be in the end-to-end notebook)

#	Missing Item	Source	Why Critical
1	**Statistical analysis tables** (ANOVA, effect sizes, confidence intervals)	NB 08	Required by project outline §11 for "statistically defensible comparison"
2	**CBD robustness comparison** (CBD vs non-CBD RT, coverage)	NB 09	Required by project outline §3 and §11: "Manhattan-wide results with CBD-focused robustness"
3	**Production experiment results** (Exp 1-4 summaries)	NB 07	The 810-simulation production run results are the primary evidence base
4	**Figures not saved to disk**	—	None of the 8 visualizations are saved to `results/figures/`, making them non-reproducible outside the notebook
5	**Queue/utilization metrics**	NB 07, scripts	Project outline §2 lists "queue length" and "unit utilization" as key MOEs
6	**Service-level metric** (% incidents within 8-min threshold)	NB 07	Project outline §2 requires "percentage of incidents served within a defined threshold"

Important Gaps (useful but not essential)

#	Missing Item	Source	Why Important
7	**Crash density heatmap** (spatial map)	NB 02	Visually demonstrates demand concentration; strong for presentations
8	**Firehouse locations map**	NB 02	Shows candidate staging sites on map
9	**Input model fit diagnostics** (Poisson/ NHPP)	NB 03	Demonstrates demand model credibility
10	**Allocation comparison visualization** (which firehouses get units)	NB 05	Shows spatial allocation differences between policies
11	**Response time distribution by policy** (histograms/violin)	NB 07	Richer than just mean RT; shows distributional shape
12	**Seasonal/monthly patterns**	NB 02	Shows temporal demand structure beyond hourly/DoW
13	**Capacity sensitivity analysis**	Scripts	Shows cap=2 is sufficient

Optional Gaps (detailed analysis, fine in specialized notebooks)

#	Missing Item	Source	Why Optional
14	Service time distribution visualization	NB 04	Implementation detail
15	ToD speed factor visualization	NB 04	Implementation detail
16	NHPP arrival demo	NB 04	Implementation detail
17	Detailed ANOVA assumption checks (Q-Q, Levene)	NB 08	Statistical detail
18	P0 spatial stratification analysis (latitude/grid/maximin)	Scripts	Design validation detail
19	Manhattan distance comparison	Scripts	Robustness check detail
20	Seasonal decomposition	NB 02	Extended EDA
21	Queue heatmap & detailed queue analysis	Scripts	Advanced diagnostics
22	Publication-quality figure generation	Scripts	Separate publication pipeline
23	Project summary dashboard	Scripts	Standalone summary artifact

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Section D: Recommendations for Enhancement

Priority 1 — Critical (should be added)

D1. Add CBD Robustness Section (Section 6.4 or new Section 6.5)

- **What:** Run P0 and P2 simulations at K=20 with CBD/non-CBD RT breakdown
- **Content:** Table showing CBD RT vs non-CBD RT for each policy; bar chart comparison
- **Data source:** Can reuse existing simulation results or run lightweight CBD analysis
- **Effort:** ~30 lines of code + 1 visualization
- **Justification:** Project outline explicitly requires CBD robustness comparison

D2. Add Statistical Summary Table (Section 7 enhancement)

- **What:** Include a compact ANOVA + effect size summary for the key policy comparison
- **Content:** F-statistic, p-value, Cohen's d for P0 vs P2 at K=20
- **Data source:** Can compute from the simulation replication data already generated in the notebook
- **Effort:** ~20 lines of code + 1 formatted table
- **Justification:** Required for "statistically defensible comparison"

D3. Add Production Experiment Summary (new Section 6.5 or enhanced Section 7)

- **What:** Load and display the production experiment results (4 experiments, 810 runs)
- **Content:** Summary table of all 4 experiments with key metrics; reference to full results
- **Data source:** Load from `results/simulation/production/experiment_summary.csv`
- **Effort:** ~15 lines of code
- **Justification:** These are the primary evidence for the recommendation

D4. Save All Figures to Disk

- **What:** Add `savefig()` calls for each visualization produced
- **Content:** Save to `results/figures/e2e_*.png` with consistent naming
- **Effort:** ~10 lines (one per figure)
- **Justification:** Reproducibility; figures should be available outside notebook execution

D5. Add Queue/Utilization Metrics

- **What:** Add queue length and utilization statistics to the simulation summary
- **Content:** Print mean queue length and unit utilization alongside response time
- **Effort:** ~10 lines (data already in simulation output)
- **Justification:** Listed as key MOEs in the project outline

Priority 2 — Important (recommended)

D6. Add Crash Density Heatmap (Section 3 enhancement)

- **What:** Add a spatial heatmap of crash density across Manhattan precincts
- **Effort:** ~15 lines using existing precinct GeoDataFrame
- **Justification:** Strong visual for spatial demand understanding

D7. Add Allocation Comparison Visualization (Section 4 enhancement)

- **What:** Show which firehouses receive units under P0 vs P2 (side-by-side bar or map)
- **Effort:** ~20 lines
- **Justification:** Makes the optimization tangible — what actually changes

D8. Add Input Model Summary (Section 3 enhancement)

- **What:** Brief NHPP model summary with hourly rates plot and precinct rates
- **Effort:** ~15 lines
- **Justification:** Closes the "input modeling" gap in the end-to-end narrative

Priority 3 — Nice to Have

D9. Add Response Time Distribution Plot

- **What:** Histogram or violin plot of response times by policy (not just box plot)
- **Effort:** ~10 lines

D10. Add Monthly/Seasonal Pattern

- **What:** Add monthly crash volume chart to EDA section
- **Effort:** ~10 lines

Summary Score Card

Category	End-to-End NB	Full Project	Coverage
Inline visualizations	8-9	60+ figures	~14%
Saved figures	0	60	0%
Summary tables	6 (printed)	18+ CSV tables	~33%
Statistical analysis	None	ANOVA, effect sizes, CIs, post-hoc	0%
CBD robustness	None	Full CBD analysis	0%
Queue/utilization metrics	None	Queue stats, heat-maps	0%
Verification & valida-tion	Complete (7/7 tests)	Same	100%
Optimization compar-ison	Complete (5 policies × 4 K values)	Same + more K val-ues	~80%

Estimated Enhancement Effort

Priority	Items	Estimated Lines of Code	Estimated Runtime Impact
Critical (P1)	D1–D5	~85 lines	+30–60s (CBD simu-lation)
Important (P2)	D6–D8	~50 lines	Negligible
Nice-to-have (P3)	D9–D10	~20 lines	Negligible
Total	**10 items**	**~155 lines**	**+30–60s**

Conclusion

The end-to-end notebook covers the **core workflow** well — data loading, EDA, optimization, simula-tion V&V, and results summary. However, it has **significant gaps** in three areas required by the project outline:

1. **No statistical analysis** (ANOVA, effect sizes, confidence intervals)

2. **No CBD robustness comparison** (explicitly required)
3. **No production experiment results** (the 810-run evidence base is not referenced)
4. **No figures saved to disk** (reproducibility concern)

The notebook currently functions as a **demonstration workflow** rather than a **comprehensive results showcase**. Adding the 5 critical items (~85 lines of code) would transform it into a truly end-to-end deliverable that matches the project outline's requirements.